D. Yet Another Array Problem

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: brute force, implementation, math, number theory |
https://codeforces.com/problemset/problem/2167/D

You are given an integer n and an array a of length n.

Find the **smallest** integer x ($2 \le x \le 10^{18}$) such that there exists an index i ($1 \le i \le n$) with $gcd^*(a_i, x) = 1$. If no such x exists within the range $[2,10^{18}]$, output -1.

*gcd(x, y) denotes the greatest common divisor (GCD) of integers x and y.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

Each of the following t test cases consists of two lines:

The first line contains a single integer n ($1 \le n \le 10^5$) — the length of the array.

The second line contains n space-separated integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^{18}$).

It is guaranteed that the total sum of n across all test cases does not exceed 10^5.

Output

For each test case, output a single integer: the smallest x ($2 \le x \le 10^{18}$) such that there exists an index i with gcd(a_i, x) = 1. If there is no such x in the range $[2,10^{18}]$, print -1.

Example

Input
4 1 1 4 6 6 12 12 3 24 120 210 4 2 4 6 10

Output
2 5 5 3

Note

In the first test case, gcd(2,1)=1, which is the smallest number satisfying the condition.

In the second test case:

- gcd(2,6)=2, gcd(2,12)=2, so 2 cannot be the answer.
- gcd(3,6)=3, gcd(3,12)=3, so 3 cannot be the answer.
- gcd(4,6)=2, gcd(4,12)=4, so 4 cannot be the answer.
- gcd(5,6)=1, so the answer is 5.

In the third test case:

- gcd(2,24)=2, gcd(2,120)=2, gcd(2,210)=2, so 2 cannot be the answer.
- gcd(3,24)=3, gcd(3,120)=3, gcd(3,210)=3, so 3 cannot be the answer.
- gcd(4,24)=4, gcd(4,120)=4, gcd(4,210)=2, so 4 cannot be the answer.
- gcd(5,24)=1, so the answer is 5.

In the fourth test case:

- gcd(2,2)=2, gcd(2,4)=2, gcd(2,6)=2, gcd(2,10)=2, so 2 cannot be the answer.
- gcd(3,2)=1, so the answer is 3.